

LANGUAGE: C PROGRAMMING LEVEL:INTERMEDIATE-ADVANCED

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
Day 1	Revision of C Basics	C Program Structure, Compilation Process, Data Types, Variables, Constants, Operators, Conditional Statements, Loops, Functions, Arrays, Strings, Searching & Sorting Basics	LeetCode: Two Sum, Maximum Subarray, Valid Anagram HackerRank: Arrays DS, C Tutorial Basics	Student Result Analysis System
Day 2	Pointers in C – Fundamentals	Pointers in C, Pointer Declaration & Initialization, Dereference Pointer, NULL Pointer, void Pointer, Pointer Arithmetic, Pointers and Arrays, Pointer vs Array, Pointer to Array, Array of Pointers, Double Pointers	LeetCode: Move Zeroes, Merge Sorted Array, Remove Duplicates from Sorted Array HackerRank: Pointer in C, Array Reversal	Dynamic Memory-Based Student Record System
Day 3	Advanced Pointer Concepts	Pointer to Pointer, Chain of Pointers, Character Pointers and Functions, Passing Pointers to Functions, Return Pointer from Functions, Function Pointers, Array of Function Pointers, Pointers to Structures, Restrict Keyword, Const Pointer & Pointer to Const, Dangling Pointers	LeetCode: Linked List Cycle, Reverse String, Intersection of Two Arrays HackerRank: Function Pointers, Structs in C	Menu Driven Banking System using Function Pointers
Day 4	Structures, Unions & Memory Alignment	Structures, Nested Structures, Unions, Enumeration, typedef, Structure Padding, Structure Alignment, Data Packing, Bit Fields	LeetCode: Richest Customer Wealth, Find Pivot Index HackerRank: Small Triangles Large Triangles	Employee Management System
Day 5	Memory Management in C	Memory Layout, Stack vs Heap, Memory Address, Storage Classes, malloc(), calloc(), realloc(), free(), Dynamic Array Resizing, Memory Leaks, Debugging Memory Issues	LeetCode: Contains Duplicate, Missing Number, Climbing Stairs HackerRank: Dynamic Array	Dynamic Inventory Management System
Day 6	File Handling in C	File Handling, Input & Output, File Operations, Reading/Writing Structures, Formatted Output, getc(), getchar(), getch(), getche(), Binary Files vs Text Files	LeetCode: Valid Sudoku, Matrix Diagonal Sum HackerRank: File I/O	Student Database using Files
Day 7	Preprocessors & Macros	Preprocessors, Macros, Header Files, Custom Header Files, Preprocessor Operators, Working of Preprocessor, Pragmas, Conditional Compilation	LeetCode: Power of Two, Reverse Integer HackerRank: C Preprocessor Exercises	Configurable Logging Framework
Day 8	Advanced Functions & Variable Arguments	Variadic Functions, Variable Arguments, Recursive Functions, Inline Functions, Static Keyword, Command Line Arguments	LeetCode: Fibonacci Number, Pascal Triangle HackerRank: Functions in C	Command Line Utility Tool
Day 9	Error Handling & System Programming	Error Handling, perror(), errno, Assertions, goto for cleanup, Command Execution using system(), Math Functions, Random Number Generation	LeetCode: Single Number, Happy Number HackerRank: Error Handling Programs	Linux-Based Automation Utility

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Day 10	Multithreading & Signals in C	Introduction to Multithreading, pthread Basics, Thread Creation & Synchronization, Mutex, Race Conditions, Signals in C, Signal Handling	LeetCode: Print in Order, FizzBuzz Multithreaded HackerRank: Threading Basics	Multi-threaded Log Processing System
Day 11	Socket Programming & System Concepts	Socket Programming Basics, TCP/IP Fundamentals, Client-Server Architecture, Socket APIs, Communication using Sockets, Process vs Thread	LeetCode: Number of Islands, Flood Fill HackerRank: Networking Basics	Client-Server Chat Application
Day 12	Advanced Interview Problem Solving	Time Complexity, Space Complexity, Sliding Window, Hashing, Searching & Sorting Optimization, Debugging, Output Prediction, Corporate Coding Patterns	LeetCode: Two Sum, Maximum Subarray, Valid Parentheses, Merge Intervals, Best Time to Buy & Sell Stock, Binary Search HackerRank: Sparse Arrays, Balanced Brackets, Left Rotation, Problem Solving (Easy-Medium)	Campus Interview Coding Round Simulation
Day 13	Advanced Coding + RDBMS Concepts	First Half – Coding Practice: Mock Interview Coding Round, Optimization Techniques, Advanced MCQs, Debugging Round, Code Review Concepts. Second Half – RDBMS Concepts: DBMS vs RDBMS, ER Model, Keys, Normalization, SQL Basics, SELECT, INSERT, UPDATE, DELETE, Joins, Aggregate Functions	LeetCode: Rotate Array, Contains Duplicate, Valid Anagram, Climbing Stairs HackerRank: SQL Basic Select, Weather Observation Station, Dynamic Array	Student Database Management System
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Overview of Cloud Computing, Types of Cloud Services (IaaS, PaaS, SaaS), Cloud Deployment Models, Virtualization, Git & GitHub Basics, clone/push/pull/merge conflicts, Docker Basics, Docker Commands, Docker Images, Docker Networking & Volumes, Docker Compose, Kubernetes Architecture & Components, K8S Commands, Deploying Applications on K8S, K8S Networking & Service Discovery, Introduction to DevOps, CI/CD Pipelines, Infrastructure as Code (IaC), GitHub Actions/Jenkins Basics	LeetCode: Single Number, Move Zeroes, Merge Intervals HackerRank: Linux Shell Basics, Git Tutorials, 30 Days of Code	End-to-End DevOps Deployment Pipeline using GitHub + Docker + Kubernetes